

# CCSW432 – IoT

Department of Computer Science

College of Computer Science and Engineering

Project

Smart Vehicle Anti-Theft and Monitoring System

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Prepared by:

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## 1. Project Aim

The main goal of this smart vehicle anti-theft and monitoring system is to make vehicles safer and protect them from unauthorized individuals. By using an ESP32 microcontroller, the system helps prevent people from taking vehicles that do not belong to them.

## 2. Project Description

This project introduces a Smart Vehicle Anti-Theft and Monitoring System based on the ESP32 microcontroller. The project uses a variety of security measures to prevent unauthorized individuals from accessing cars and other vehicles. In addition, the system has a relay module to turn on the car lock, a buzzer to sound alerts, a tilt sensor to identify suspicious movement, and a keypad for password authentication. Additionally, the vehicle's location can be tracked and alerts are transmitted by GPS and GSM modules. The ESP32 additionally has a web server that allows for remote system control and monitoring.

## 3. System Working

- 1.The ESP32 initializes all the (connected) hardware components.
- 2.The system is armed and vehicle locked.
- 3.The user inputs a password via the keypad.
- 4.If password is correct, re lock opens.
- 5.In case input password is incorrect multiple times, the theft alert is triggered.
- 6.Tilt sensor is continuously monitoring vehicle movement.
- 7.In case of movement of vehicle when armed,buzzer alarm is triggered.
- 8.GPS is used to determine the vehicle location.
- 9.GSM/WiFi is used for remote monitoring and alerts.

#### 4. Hardware Components

- ESP32 Development Board (ESP32-WROOM-32)
- GSM/GPRS Module (SIM800L)
- GPS Module (GY-NEO6MV2 – NEO-6M)
- Tilt/Vibration Sensor (SW-420)
- 5V Single Channel Relay Module
- 4×4 Matrix Keypad
- Buzzer Module
- Breadboard (DIY Prototype Board)
- Jumper Wires (Male–Male / Male–Female)
- 18650 Battery Charging Module (5V)
- Lithium-Ion Batteries (18650 – 2600 mAh)

#### 5. Software Components

- Arduino IDE
- Embedded C / Arduino Programming Language
- WiFi.h Library
- AsyncTCP.h Library
- ESPAsyncWebServer.h Library
- Keypad.h Library
- SPIFFS.h Library

#### 6. Applications

- Vehicle anti-theft systems
- Smart transportation security
- Fleet monitoring systems
- IoT-based security applications

#### 7. Conclusion

The Smart Vehicle Anti-Theft and Monitoring System clarifies the potential of utilizing IoT and embedded techniques for vehicle security. This concludes that authentication, sensors and communication, and a real-time monitoring system has been based on an efficient and scalable system for real-world applications.